FINITE TYPE DESCENDS ALONG A FAITHFULLY FLAT FINITELY PRESENTED ALGEBRA MAP

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays the approximation diagram and the isomorphism identifying the two base changed maps; both lies inside the counted span and are reproduced verbatim.

Editorial note. The argument is a lemma about ring maps proved inside this chapter of scheme theory, and the label reflects the role it plays in fppf locality on the source; the card follows the statement.

Editorial note. The counted spans carry 1 navigation stub; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Approximation of a pair of ring maps by a finitely presented model. Algebra, Lemma algebra-lemma-descend-faithfully-flat-finite-presentation, invoked at proof lines 4577-4579. Produces the commutative diagram with the finite presentation and faithful flatness properties used in the rest of the proof.
- Surjectivity descends along a faithfully flat ring map. Standard, used at proof lines 4593-4598. Turns surjectivity of the base change of the map from A_0 to A into surjectivity of the map itself.
- A composition of ring maps is of finite type when the first is of finite presentation and the second is surjective. Standard, used at proof lines 4590-4592 and 4598. Concludes finite type for R to A.

Source reference. Original lemma 034Y: sources/stacks-chow/tags.txt, source lines 4109–4109. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $R \to A \to B$ be ring maps. Assume $R \to B$ is of finite type and $A \to B$ faithfully flat and of finite presentation. Then $R \to A$ is of finite type.*

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Proof. By Algebra, Lemma 034Y there exists a commutative diagram

$$\begin{array}{ccccc} R & \longrightarrow & A_0 & \longrightarrow & B_0 \\ \| & & \downarrow & & \downarrow \\ R & \longrightarrow & A & \longrightarrow & B \end{array}$$

with $R \to A_0$ of finite presentation, $A_0 \to B_0$ faithfully flat of finite presentation and $B = A \otimes_{A_0} B_0$. Since $R \to B$ is of finite type by assumption, we may add some elements to A_0 and assume that the map $B_0 \to B$ is surjective! In this case, since $A_0 \to B_0$ is faithfully flat, we see that as

$$(A_0 \to A) \otimes_{A_0} B_0 \cong (B_0 \to B)$$

is surjective, also $A_0 \to A$ is surjective. Hence we win. $\square$